

Diseño de un controlador para un sistema de segundo orden.

- Ecuaciones Principales.

$$V_e(t) = R I_1(t) + L \frac{d[I_1(t) - I_2(t)]}{dt} + R [I_1(t) - I_2(t)]$$

$$L \frac{d[I_1(t) - I_2(t)]}{dt} + R [I_1(t) - I_2(t)] = R I_2(t) + R I_2 + \frac{1}{C} \int I_2(t) dt$$

$$V_s(t) = R I_2(t) + \frac{1}{C} \int I_2(t) dt$$

- Transformada de Laplace.

$$V_e(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$

$$L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{Cs}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{Cs}$$

- Procedimiento Algebraico.

$$V_e(s) = (Ls + 2R) I_1(s) - (Ls + R) I_2(s)$$

$$I_1(s) = \frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s)$$

$$V_s(s) = (R + \frac{1}{Cs}) I_2(s)$$

$$\frac{V_s(s)}{V_e(s)} = \frac{(R + \frac{1}{Cs}) I_2(s)}{(Ls + 2R) I_1(s) - (Ls + R) I_2(s)} = \frac{\frac{CRs + 1}{Cs} \cdot I_2(s)}{(Ls + 2R) (\frac{1}{Ls + R} (Ls + \frac{1}{Cs} + 3R) I_2(s)) - (Ls + R) I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs + 1}{Cs} \cdot I_2(s)}{(Ls + 2R) \left(\frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} I_2(s) \right) - (Ls + R) I_2(s)} = \frac{\frac{CRs + 1}{Cs} \cdot I_2(s)}{\left((Ls + 2R) \frac{CLs^2 + 3CRs + 1}{CLs^2 + CRs} - (Ls + R) \right) I_2(s)}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{CRs + 1}{Cs}}{\frac{3CLRs^2 + (5CR^2 + L)s + 2R}{CLs^2 + CRs}} = \frac{(CRs + 1)(Ls + R)}{3CLRs^2 + (5CR^2 + L)s + 2R}$$

$$\frac{V_s(s)}{V_e(s)} = \frac{CLAs^2 + (CR^2 + L)s + R}{3CLRs^2 + (5CR^2 + L)s + 2R} \quad \text{Función de Transferencia}$$

- Error en estado estacionario.

$$\lim_{s \rightarrow 0} s R(s) [1 - C(s)] \Rightarrow \frac{0 \cdot R(0)^2 + (CE^2 + L)(0) + R}{3CLR(0) + (5CR^2 + L)(0) + 2R} = \frac{R}{2R} = \frac{1}{2} = 0.5$$

$$e_{ss} = 0.5 = 50\%$$

- Estabilidad en lazo abierto.

$$R = 5k\Omega$$

$$L = 10^{-4}H$$

$$C = 470^{-6}F$$

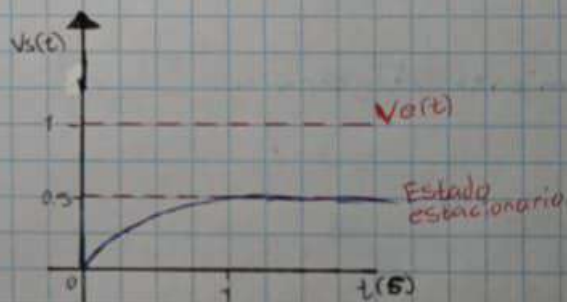
$$3CLR s^2 + (5CR^2 + L)s + 2R$$

$$S_{1,2} = \frac{-(5(470E^{-6})(5E^3)^2 + (10E^{-6})) \pm \sqrt{(5(470E^{-6})(5E^3)^2 + (10E^{-6}))^2 - 4(3(470E^{-6})(10E^{-6})(5E^3))}}{2(3(470E^{-6})(10E^{-6})(5E^3))}$$

$$S_1 = -414.947672E^6$$

$$S_2 = -418.38566E^6$$

Ambos polos son negativos, por lo tanto, se obtiene una Respuesta Sobreamortiguada en un sistema estable.



- Modelo de Ecuaciones Integro/Diferencial.

El modelo se expresa despejando las variables dependientes, es decir, las corrientes:

$$V_c(t) = 2RI_1(t) + L \frac{d[I_1(t) - I_2(t)]}{dt} - RI_2(t)$$

$$I_1(t) = [V_c(t) + RI_2(t) - L \frac{d[I_1(t) - I_2(t)]}{dt}] \cdot \frac{1}{2R}$$

$$L \cdot \frac{d[I_1(t) - I_2(t)]}{dt} + RI_1(t) = RI_2(t) + RI_2(t) + RI_2(t) + \frac{1}{C} \int I_2(t) dt$$

$$I_2(t) = [L \cdot \frac{d[I_1(t) - I_2(t)]}{dt} + RI_2(t) + \frac{1}{C} \int I_2(t) dt] \cdot \frac{1}{3R}$$

$$I_1(t) = [V_c(t) + RI_2(t) - L \frac{d[I_1(t) - I_2(t)]}{dt}] \cdot \frac{1}{2R}$$

$$I_2(t) = [L \cdot \frac{d[I_1(t) - I_2(t)]}{dt} + RI_1(t) - \frac{1}{C} \int I_2(t) dt] \cdot \frac{1}{3R}$$

$$V_c(t) = RI_2(t) + \frac{1}{C} \int I_2(t) dt$$